Kerollos Mansour

Software AI Engineer

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Summary

Software AI Engineer who studied Computer Science and Artificial Intelligence, with a track record of designing, building, and shipping production-grade software systems end-to-end — from backend architecture and APIs to intelligent conversational applications and advanced retrieval and reasoning platforms. My passion is creating AI products and working hands-on to bring them to life, combining solid software engineering practices with strong foundations in machine learning and deep learning to integrate and deploy AI models into real, working products. I care about systems that don't just work in a notebook, but hold up in production — scalable, observable, and built to last.

Technical Skills

- **Programming Languages:** Python, Java, C++, SQL.
- **Mobile Development:** Android (Java), Flutter
- **Frameworks & Libraries:** LangChain, LangGraph, RAG pipelines, hybrid retrieval (vector + BM25 + graph), PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy
- **Backend & Software Engineering:** FastAPI, Flask, Streamlit, Celery, Redis, RESTful APIs, API integration, object-oriented programming, microservices, software architecture, debugging, testing
- **LLM Providers & Local Models:** OpenAI API, Azure OpenAI, Groq, Hugging Face, Ollama
- **Deployment & MLOps:** Docker, MLflow, Keycloak (JWT/RBAC), Caddy (TLS reverse proxy), Prometheus, Grafana, Locust, model serving, model integration, deployment pipelines
- **Databases & Tools:** PostgreSQL, Qdrant, Apache AGE, MySQL, Git, GitHub, Postman

Soft Skills

- Problem-Solving & Analytical Thinking
- Critical Thinking & Research
- Teamwork & Collaboration
- Adaptability & Continuous Learning
- Time Management & Adaptability
- Communication Skills
- Creativity & Innovation

Experience

IBEX Solutions | Software AI Engineer **Sep 2024 – Present**

- Designed and built IBEXA, an AI-powered customer service chatbot, integrating LLMs into backend services end-to-end (requirements, architecture, development, testing, deployment) and deploying it to production, improving scalability and maintainability.
- Developed and deployed REST API endpoints and third-party integrations powering IBEXA's conversational workflows, reducing integration defects by 25% and improving release stability.
- Optimized database queries and data models behind IBEXA's retrieval and response pipeline, cutting average API response time by 30% and improving throughput by 20% under peak load.
- Automated testing and improved observability (logs/metrics/alerts) for the AI chatbot pipeline, increasing test coverage to 70% and reducing production incidents by 35%.

- Collaborated with product, QA, and engineering in Agile sprints to ship IBEXA features and fixes, delivering +15 features/bug fixes per quarter with 90% on-time completion.

Fixed Solutions | Intern Software Engineer **Jun 2026 – Jul 2026**

- Engineered a multi-user, multi-domain RAG system with hybrid retrieval combining dense vector search (Qdrant), BM25 sparse search (PostgreSQL), and graph traversal (Apache AGE), fused via Reciprocal Rank Fusion and cross-encoder reranking.
- Architected a unified FastAPI monolith consolidating 5 backend microservices (domain, ingestion, retrieval, generation, evaluation) into a single deployable process, cutting inter-service network overhead and simplifying deployment.
- Built an asynchronous document ingestion pipeline (Celery + Redis) with OCR (PaddleOCR, Surya fallback) and semantic chunking, automating extraction and dual-indexing of PDF, DOCX, CSV, and image files into Qdrant and PostgreSQL.
- Secured the platform end-to-end with JWT authentication and role-based access control (RBAC) via Keycloak, and TLS-terminated Caddy reverse proxies across all service endpoints.
- Extracted entities and relations (GLiNER + LLM) to construct a knowledge graph layer (Apache AGE), extending retrieval beyond vector/BM25 to answer relational queries.
- Instrumented the system with Prometheus and Grafana dashboards and load-tested it with Locust under concurrent query traffic, tuning DB connection pools, Redis memory, and Uvicorn worker counts to resolve throughput bottlenecks.

National Telecommunication Institute (NTI) | Agentic AI Trainee **Jan 2025 – Apr 2026**

- Engineered and deployed production-grade agentic AI systems powered by LLMs through an intensive 288-hour advanced technical program.
- Developed LLM-powered agentic applications for real-world use cases, focusing on orchestration, retrieval, and response quality.
- Implemented advanced RAG architectures with memory and reranking components to deliver grounded and context-aware outputs.
- Built autonomous and collaborative agent systems with LangChain and LangGraph to support complex task execution.

Education

Bachelor of Computer Science and Artificial Intelligence - ***Cairo University*** **Sep 2021 – Jul 2025**

As a Computer Science student specializing in Artificial Intelligence, I have:

- Built a strong foundation in software development, data structures, and object-oriented programming.
- Explored core AI disciplines such as machine learning, deep learning, and natural language processing.
- Developed hands-on projects using the **PyData stack** (Pandas, NumPy, scikit-learn) for data analysis and ML applications.
- Gained practical experience with the **Deep Learning stack** (TensorFlow, Keras) for building and deploying neural networks.
- Collaborative development with Git, GitHub, and agile methodologies.

Certifications

- **Supervised Machine Learning: Regression and Classification** – Coursera, Jan 2026
- **Advanced Learning Algorithms** – Coursera, Jan 2026
- **Unsupervised Learning, Recommenders, Reinforcement Learning**– Coursera, Feb 2026
- **Huawei HCIA-AI Learning Course** – Huawei, Jul 2024